



AUSTIN AMATEUR RADIO CLUB · EST. 1916

The AARC Linked Repeater System

Five sites. One conversation. Motorola MOTOTRBO DMR, linked over IP — plus open analog repeaters for everyone.

2 sites on the air · 3 sites in build · **Communication** · **Education** · **Preparedness**

Membership presentation · Presented by Chris Kacerguis, KJ5DJC · September 2026

Where the system stands today

5

Repeater sites

Planned across the Austin metro

2

On the air now

Operational and carrying traffic

3

In the build queue

Hardware secured, sites pending

\$10

Per month

Fully configured loaner DMR +
Analog UHF radio

What this gets us

- Digital talkgroups heard on every linked site — key up in Round Rock, get answered in Buda.
- Radios roam site to site automatically as you drive, with no channel changes.
- Repeater software, codeplugs, and health monitoring managed centrally
- Analog stays exactly as it is today: open to all licensed amateurs, no membership required.



Hardware standard

**Motorola MOTOTRBO
SLR 5700 and SLR 8000**

Two systems, one club

The club runs two parallel repeater services. They share sites and antennas in places, but they serve different purposes and have different access rules.



MEMBERS ONLY

DMR — linked

Motorola MOTOTRBO, TDMA digital

- All DMR sites are linked over IP into one system.
- Access requires a club-issued radio ID — members only.
- Automatic roaming between sites while mobile.
- Talkgroups separate club nets, chat, and events.
- Managed and updated centrally by the repeater team.
- Network Access Code



OPEN TO ALL

Analog — open

FM, conventional repeater operation

- Open to any licensed amateur. No membership needed.
- Not currently linked — each machine stands alone.
- Standard PL/CTCSS access, no radio ID required.
- Remains the club's public front door and net home.
- Linking analog is not part of this project.



How the sites tie together



The five-site roster

Site	Repeater	Band / mode	Linked	Status
North Austin Temp Site	Motorola SLR 5700 (50w)	UHF · DMR + analog	Yes	On the air
South Austin Temp Site	Motorola SLR 5700 (50w)	UHF · DMR + analog	Yes	On the air
Lakeway Site	Motorola SLR 5700 (50w)	VHF · DMR + analog	Planned	Waiting on Antenna
Buckman Mountain	Motorola SLR 8000 (100w)	UHF · DMR + analog	Planned	Seeking Funding
Site 5 — East / southeast	Motorola SLR 8000	UHF · DMR + analog	Planned	Survey stage



What "on the air" means

- Repeater installed, tuned, and coordinated.
- IP link established and passing traffic.
- Member codeplug published for the site.



What the pending sites need

- Waiting for location
- Waiting for equipment (Antennas are back ordered)
- Seeking Funding

RF parameters throughout this deck are working values — confirm against current coordination records before publishing to the membership.

North Austin Site

ON THE AIR



Our northern anchor, hosted on a member’s property. Highest effective height of the two live sites and the current IPSC master for the linked system.

REPEATER	Motorola SLR 5700 (UHF)	MODE	DMR Tier II, 2 timeslots
OUTPUT / INPUT	441.3875 MHz, +5 MHz offset, 103.5 Tone	COLOR CODE	CC1
TX POWER	50 W into the duplexer	TALKGROUPS	Standard TG Plan
ANTENNA	Diamond Antenna X50NA, ~50 ft AGL	FEEDLINE	LMR-400UF, lightning arrested, grounded
DUPLEXER	Bandpass/reject, factory tuned	BACKHAUL	Google Fiber + Spectrum + Starlink + FirstNet
POWER	City Power with Whole Home battery (~3 days)	LINK ROLE	Manager Site



Site notes

This site is on private residential property. Site meets criteria and is located at an ARRL Official Emergency Station. We are looking for a more permanent space. Suggestions / Help is welcome!

South Austin Site

ON THE AIR



The southern half of the live pair, also member-hosted. Fills in coverage south of the river and gives roaming radios a second site to hand off to.

REPEATER	Motorola SLR 5700 (UHF)	MODE	DMR Tier II, 2 timeslots
OUTPUT / INPUT	441.3875 MHz, +5 MHz offset, 103.5 Tone	COLOR CODE	CC1
TX POWER	50 W into the duplexer	TALKGROUPS	Standard TG Plan
ANTENNA	Diamond Antenna X50NA, ~50ft AGL	FEEDLINE	LMR-???, lightning arrested, grounded
DUPLEXER	Bandpass/reject, factory tuned	BACKHAUL	Google Fiber + Starlink
POWER	City Power + Whole Home Generator	LINK ROLE	IPSC peer



Site notes

This site is on private residential property. Site has redundant power and internet. We are looking for a more permanent space. Suggestions / Help is welcome!



What is coming next

Hardware for all three is either in hand or on order. What each site is waiting on is access, not equipment. Names below are working labels and will change once agreements are signed.

3

Lakeway Site

Baylor Scott and White

REPEATER	SLR 5700 (50w), VHF
MODE	DMR Tier II + analog
FREQ	145.180 MHz, -107.2 (Narrow)
COLOR CODE	CC1 (system-wide)
HEIGHT	Rooftop, ~150 ft AGL
BACKHAUL	Building internet, VLAN

Waiting for Antennas

4

Mt. Buckman

Central Austin

REPEATER	SLR 8000 (100w), UHF
MODE	DMR Tier II + analog
FREQ	444.100 MHz, +107.2
COLOR CODE	CC1 (system-wide)
HEIGHT	Tower space, ~1200 ft AGL target
BACKHAUL	TBD

Seeking Funding

5

Site 5

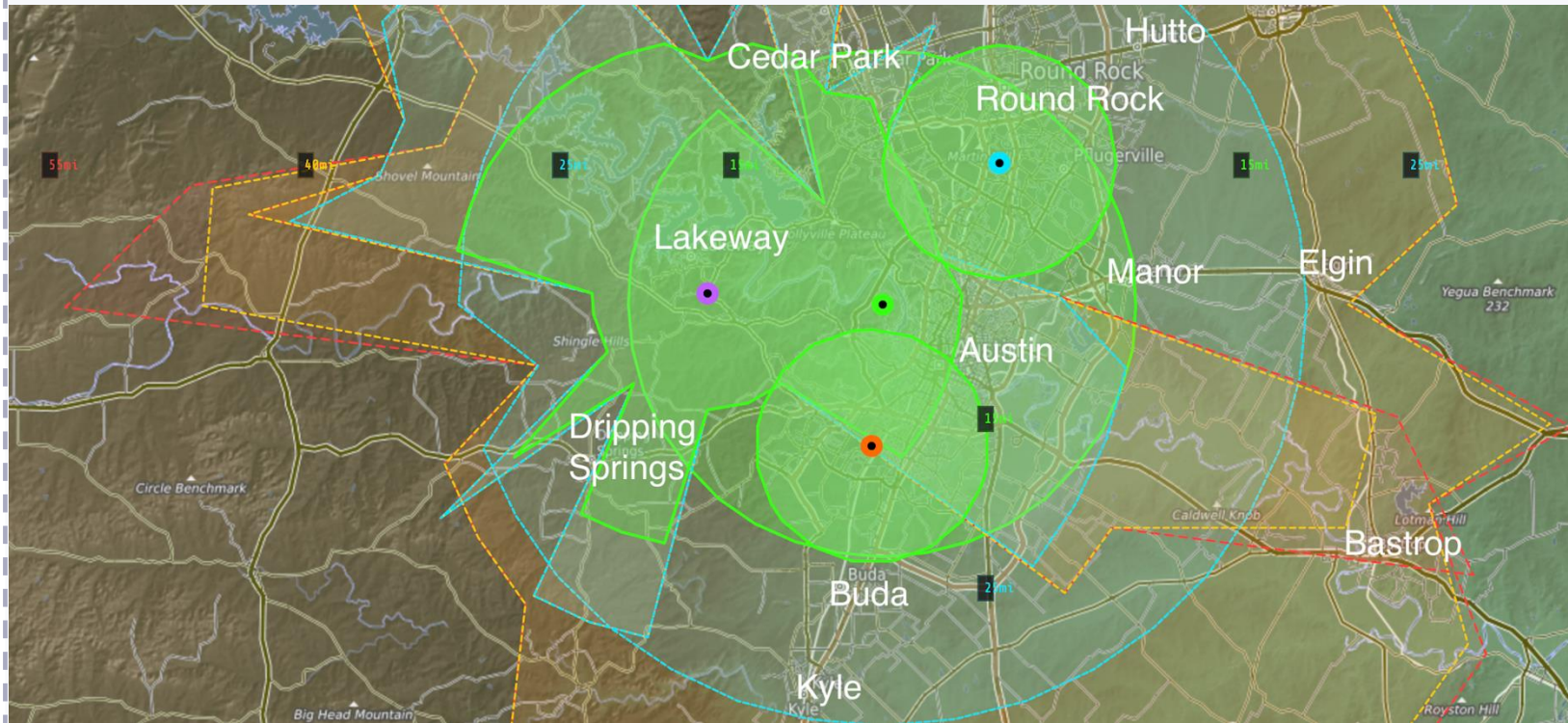
East / southeast

REPEATER	SLR 5700, UHF
MODE	DMR Tier II + analog
FREQ	Coordination pending
COLOR CODE	CC1 (system-wide)
HEIGHT	TBD
BACKHAUL	TBD

Looking for Site

All sites share one color code and talkgroups follow standard scheme. More talkgroups can be added!

Where the system reaches



Reading the map

- Inner contour: reliable handheld talk-in.
- Outer contour: mobile with a gain antenna.
- Overlap is intentional — it is what roaming needs.

Help us verify

- Predicted coverage is a model, not a promise.
- Report dead spots with location and radio type.
- Drive-test volunteers wanted before site 3 goes in.

What MOTOTRBO buys us

These are the commercial features we are actually deploying — not a catalog. Each one solves a problem a volunteer-run system has at five sites.



Roaming

Radios monitor beacons from neighbouring sites and re-register on the strongest one. Drive the metro on one channel.



OTAP

Over the Air Programming pushes codeplug updates to subscriber radios over the air. New site or new talkgroup, no cable, no meet-up. Sync'd to repeaters + Repeater Book



System management

Repeaters are configured, restarted and updated remotely. One console instead of five tower climbs.



Radio IDs & talkgroups

Every radio transmits a unique ID. Talkgroups keep nets, ragchew and event traffic on separate paths.



Monitoring

Continuous alarms on power, temperature, link state and RF forward/reflected power, with alerts to the repeater team.



No encryption

Encryption is disabled on every repeater and every issued codeplug. All traffic stays open and monitorable.

Capabilities we are deliberately leaving switched off: encryption, and any feature that would keep a licensed amateur from monitoring our traffic.

One thing we will not do



This system carries no encryption.

MOTOTRBO supports several privacy and encryption modes. Every one of them is disabled on our repeaters and left out of every codeplug we issue. Digital does not mean private — any amateur with a DMR radio can listen to us, and that is the point.

Part 97 first

Amateur rules prohibit obscuring the meaning of a communication. We stay well clear of the line.

Open to monitoring

Anyone can decode our traffic with a standard DMR radio or scanner. Members-only access is about who transmits, not who listens.

No exceptions

Not for nets, not for events, not for public service. If it needs to be private, it does not belong on the repeater.

Radio IDs and talkgroups

Every radio on the linked system carries a unique ID issued by the club. That ID identifies you on the air and gates access to the linked DMR network.

TG 1

Club

System-wide. Nets, announcements, everyday calling.

TG 2

Chat

System-wide ragchew. Keeps long QSOs off the calling group.

TG 8

Events

Public service and special events. Activated as needed.

TG 9

Local

Single site only, not linked. Handy for site testing.



Getting your ID

- Current club membership and a valid licence.
- Register your callsign with the repeater team.
- Receive your ID and the current codeplug.
- Updates arrive by OTAP after that.



Timeslot basics

Each repeater gives two independent conversations on one pair. Slot 1 carries the linked system-wide talkgroups; slot 2 is reserved for local and event traffic, so a long QSO never blocks a net.

Rent a fully configured radio

\$10

per month, per radio

Month to month. Cancel any time, no contracts, return the radio, and you are done. Simple!

The fine print

- Radios stay club property and are tracked by ID.
- Fees fund site costs and the radio pool itself.
- Damage or loss is handled case by case.
- UHF + GMRS + FRS

A DMR radio is the one real barrier to getting on the linked system. The club keeps a pool of Motorola subscriber radios, programmed and ready, and rents them to members for ten dollars a month.



Programmed before you get it

Your radio ID, all five sites, the full talkgroup plan and roaming already loaded.



Stays current automatically

Codeplug changes arrive by OTAP — you never bring it back for reprogramming.



Charger and battery included

Radio, battery, charger and belt clip. Bring a licence and a membership, nothing else.



Talking to People with Easy Entry

New hams have trouble getting on the air with lower cost radios, this will make it easy!

What You Get



Metro-wide coverage on one channel

Talk across the Austin area without hunting for a repeater or changing channels mid-drive.



Roaming that just works

Your radio re-registers on the best site by itself. Mobile operators notice this immediately.



A \$10 way in

Rent a configured radio instead of buying one and get on the air the same week you join. Easy for new hams, hopefully we end the need for more powerful (and expensive) radios for new hams.



No more programming parties

Codeplug changes are pushed over the air. Your radio stays current without a cable.



A system that stays up

Remote diagnostics mean faults get found and fixed faster, often before you notice them.



Somewhere to learn DMR

Talkgroups, IDs, slots and roaming, with club members who can help you sort out your radio.

Getting from two sites to five

PHASE 1

Prove the link

Complete

- North and South member sites on the air
- IPSC mesh established and stable
- Talkgroup plan and first codeplug published

PHASE 2

Harden and measure

In progress

- Roaming tuned and drive-tested
- Alarms and monitoring wired to the team
- Rental radio pool built out
- Codeplug updated and pushed by OTAP

PHASE 3

Add Lakeway

Next

- Get Equipment and Install

PHASE 4

Add Central and East

Planned

- Locate area for Site 5
- Full-metro coverage validated
- Analog integration reviewed at site 5

Phases 3 and 4 are gated by equipment availability. Phase 4 by Equipment and/or locations. We are looking for better North and South locations (HELP WANTED!)

What we need from the club



Site leads

Rooftop, tower or hilltop access anywhere in the metro. One introduction can unblock an entire phase.

Highest value



Drive testers

Volunteers to run the roads with a loaner radio and log where coverage holds and where it drops.

This month



Rental sign-ups

Take a radio for \$10 a month. More radios on the system is the fastest way to shake out problems.

Tonight



Build help

Antenna work, feedline, grounding, networking. Bring whatever skill you have to the next work party.

Ongoing



Talk to the repeater team

Catch us after the meeting, or reach out any time:

EMAIL

austinhams@groups.io

NET

[Weekly net, day and time]

MORE

<https://austinhams.org/repeaters/>



Questions?

Five sites. One linked digital system. Open analog for everyone, no encryption anywhere, and an easy to access system for new hams to learn and get on the air!

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